PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 2 – Choose the Data Representation

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1. Task02.md

1. Data Types Used

Field	Data Type	Reason
Floor number	int	Stores floor number
Slot number	int	Stores slot number
Vehicle number	String	Stores registration number
Entry time	LocalDateTime	Stores date and time
Duration	int	Stores parking minutes
Parking fee	double	Stores fee with decimal values

2. Array Representation

I use a two-dimensional jagged array String[][] to represent parking slots.

Each row represents a floor, and each column represents a parking slot.

It is suitable because different floors can have different numbers of slots.

Example:

String[][] slots = new String[3][];

3. Type Conversion and Casting

Parking duration is stored in minutes and converted into billable hours.

Example:

- Duration = 90 minutes
- Billable hours = 2
- Rate = Rs. 30 per hour

- Total fee = Rs. 60

Integer division can produce incorrect results. For example, 90 / 60 gives 1 instead of 1.5.

To avoid this, I use ceiling calculation:

(minutes + 59) / 60

4. Input Method

I use the Scanner class to read vehicle registration numbers, floor details and parking duration from the user.

5. Alternative Concept and Complexity Analysis

Instead of a jagged array, a one-dimensional array can be used.

Representation	Advantages	Disadvantages
Jagged array	Supports different slots per floor	Requires nested loops
1D array	Simple linear storage	Needs index calculation
ArrayList	Dynamic size	Requires collection handling

Time Complexity:

- Searching for a slot: O(N)
- Accessing a slot by index: O(1)

Space Complexity:

All representations require O(N) space for N parking slots.

Conclusion

A jagged array is the best choice because each floor can have a different number of parking slots. It is flexible and easy to manage.

2. Task02.java

```
import java.util.Scanner;

    public class Task02 {

        public static void main(String[] args) {

            Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter number of floors: ");
        int floors = sc.nextInt();
        String[][] slots = new String[floors][];
        for (int i = 0; i < floors; i++) {
            System.out.print("Enter slots in floor " + (i + 1) + ": ");
            int count = sc.nextInt();
            slots[i] = new String[count];
        }
        sc.nextLine();
        System.out.print("Enter vehicle number: ");
        String vehicleNo = sc.nextLine();
        boolean allocated = false;
        for (int i = 0; i < floors; i++) {
            for (int j = 0; j < slots[i].length; j++) {
                if (slots[i][j] == null) {
                    slots[i][j] = vehicleNo;
                    System.out.println("Vehicle allocated!");
                    System.out.println("Floor: " + (i + 1));
                    System.out.println("Slot: " + (j + 1));
                    allocated = true;
                    break;
                }
            }
            if (allocated) {
                break;
```

```
}
        }

        System.out.print("Enter parking duration in minutes: ");
        int minutes = sc.nextInt();

        int hours = (minutes + 59) / 60;
        double rate = 30.0;
        double fee = hours * rate;

        System.out.println("Parking Duration: " + minutes + " minutes");
        System.out.println("Billable Hours: " + hours);
        System.out.println("Parking Fee: Rs. " + fee);

        if (!allocated) {
            System.out.println("No parking slot available!");
        }

        sc.close();
    }
}
```

3. Sample Input

```
Enter number of floors: 3
Enter slots in floor 1: 2
Enter slots in floor 2: 3
Enter slots in floor 3: 2
Enter vehicle number: TN01AB1234
```

Enter parking duration in minutes: 90

3. Expected Output

